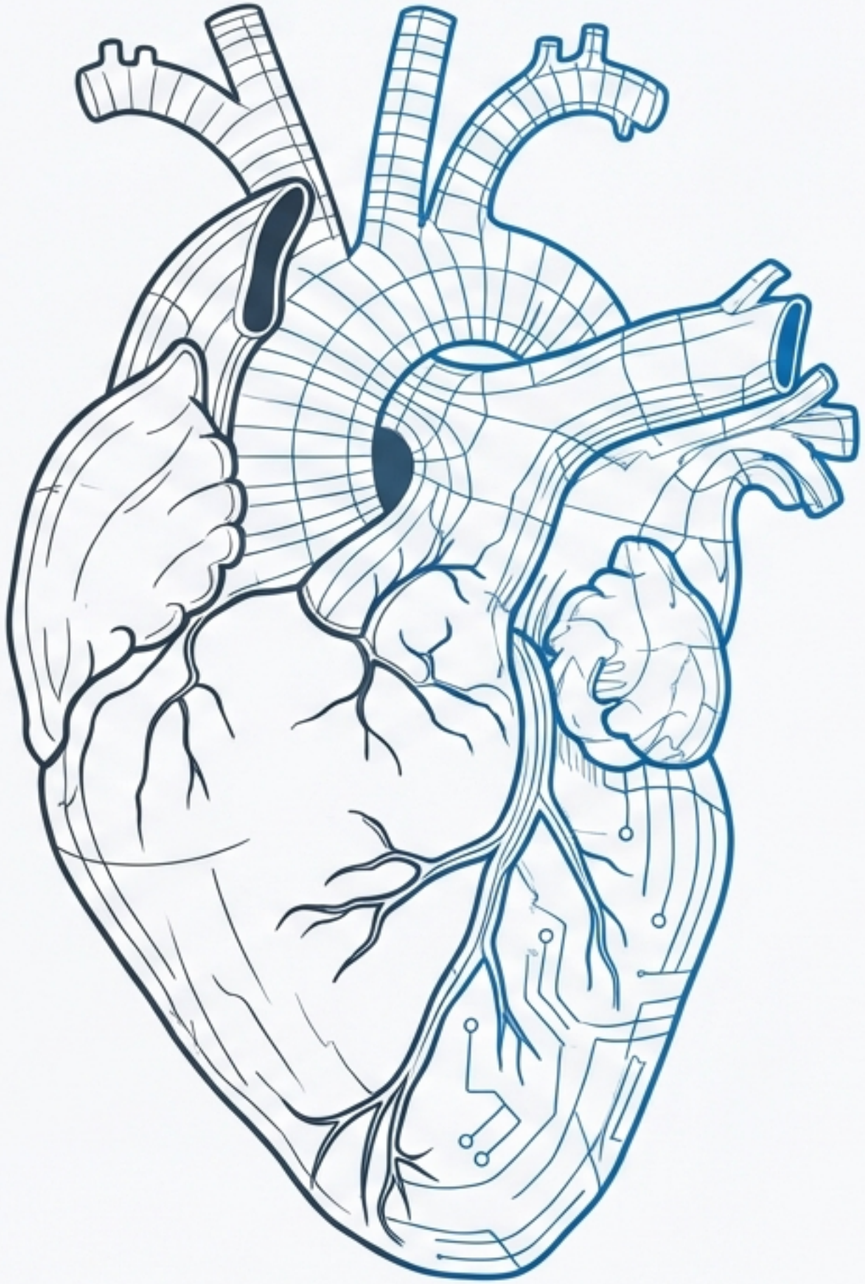
RIET
Rajadhani Institute of Engineering and Technology

Chatbot to Chat with Medical Books using Retrieval-Augmented Generation Model

Motto: MedBuddy: Grounded AI for Evidence-Based Medical Education



GROUNDED AI | PROJECT TEAM (RIE23CS)

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Project Team & Guide

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Contents

1. Abstract
2. Introduction
3. Literature Survey
4. Problem Identification
5. Proposed System
6. System and Software Requirements
7. Architecture
8. Algorithm
9. Project Plan
10. Future Scope
11. Conclusion
12. References

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Abstract

The Core Technology: This project, MedBuddy, is a digital study assistant designed to help medical students. It uses a technology called **Retrieval-Augmented Generation (RAG)**. In simple terms, RAG is a method where the AI searches for the correct answer in a trusted textbook *before* responding, rather than just guessing. We built this using **IBM Watsonx APIs**, which provides enterprise-grade, secure Artificial Intelligence tools, replacing less secure public tools.

Technical Implementation: To build this, we used the **Python 3.12** programming language managed by a tool called **Miniconda**, which ensures the software runs smoothly on any computer. We used a **'Lean AI'** strategy, meaning we optimized the heavy software libraries (like PyTorch) to run efficiently on a standard laptop **CPU** without needing expensive supercomputers.

User Experience: The student interacts with a simple website built with **Streamlit**. Behind the scenes, we use Industry-Standard **Secrets Management** to keep the API keys (passwords that let the software talk to IBM's servers) safe and hidden. This ensures the project meets the strict security standards required for handling medical data.

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Introduction

The Project Goal

MedBuddy is developed as part of the CSD 334 Mini Project curriculum. The main goal is to solve the “Disk Space Trap” and “Hallucination” problems in AI. “Hallucination” is when an AI makes up false information. MedBuddy prevents this by using trusted medical books as its only source of truth.

Why IBM Watsonx?

Unlike standard chatbots, we switched to IBM Watsonx. This is an “Enterprise-Grade” platform, which means it is designed for businesses and professional use. It allows us to use specific AI models (like the Granite-13b-chat) that are optimized to follow instructions carefully and provide concise, accurate answers from the text provided.

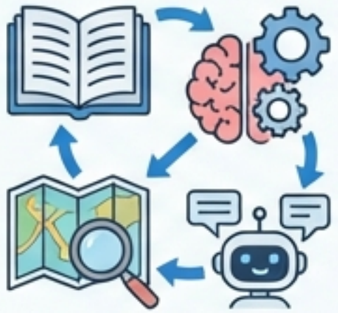


MedBuddy

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Introduction: Methodology & Datasets

The ‘Open Book’ Method



Think of MedBuddy like a student taking an open-book exam.

- **Ingestion:** It reads the textbook (PDF).
- **Indexing:** It organizes the information into a mathematical map (**Vector Database**).
- **Retrieval:** When asked a question, it finds the exact page.
- **Generation:** It writes an answer based *only* on that page.

Core Datasets Used



- **PubMed:** A massive digital library of biomedical literature from the National Institutes of Health. It is used to teach the AI general medical language.



- **Meditrin:** A specialized dataset containing specific medical definitions and clinical instructions, ensuring the AI understands doctors’ terminology.

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Literature Survey (Part 1)

Sl No.	Title [Year]	Author	Proposed Method	Disadvantages
1	Exploring AI Chatbots in Public Health [2023]	Francesco Baglivo et al.	A feasibility study on using chatbots for vaccination training.	Requires heavy professional verification to avoid the risk of spreading misinformation.
2	The Role of ChatGPT in Medical Education [2023]	Gunther Eysenbach	Investigated the capabilities of large language models like ChatGPT in clinical settings.	Raised significant concerns regarding content authenticity and the risk of 'hallucinations' (fake facts).
3	Ontology Based Personal Assistive Application [2023]	N. V. Rohil & A. Nair	Used semantic reasoning (understanding meaning) to help elderly people with daily tasks.	Focused on personal daily activity logs rather than academic medical education.

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Literature Survey (Part 2)

Sl No.	Title [Year]	Author	Proposed Method	Disadvantages
4	Farm-Haystack QA System [2022]	R. S. Nambiar et al.	Integrated Natural Language Processing (NLP) to answer pandemic-related health questions.	Requires heavy 'fine-tuning' (training) for each specific topic and lacks broad clinical utility.
5	HR based Chatbot using Deep Neural Network [2022]	P. C. Nair et al.	Used Deep Learning and the Adam optimizer to help with Human Resources communication.	Designed specifically for HR tasks; lacks the specialized medical knowledge required for healthcare.
6	AI Chatbot for Nursing Students [2022]	Jeong-Won Han et al.	Developed a chatbot program to improve nursing skills for electronic fetal monitoring.	Restricted to a single nursing skill focus; cannot answer general medical retrieval questions.

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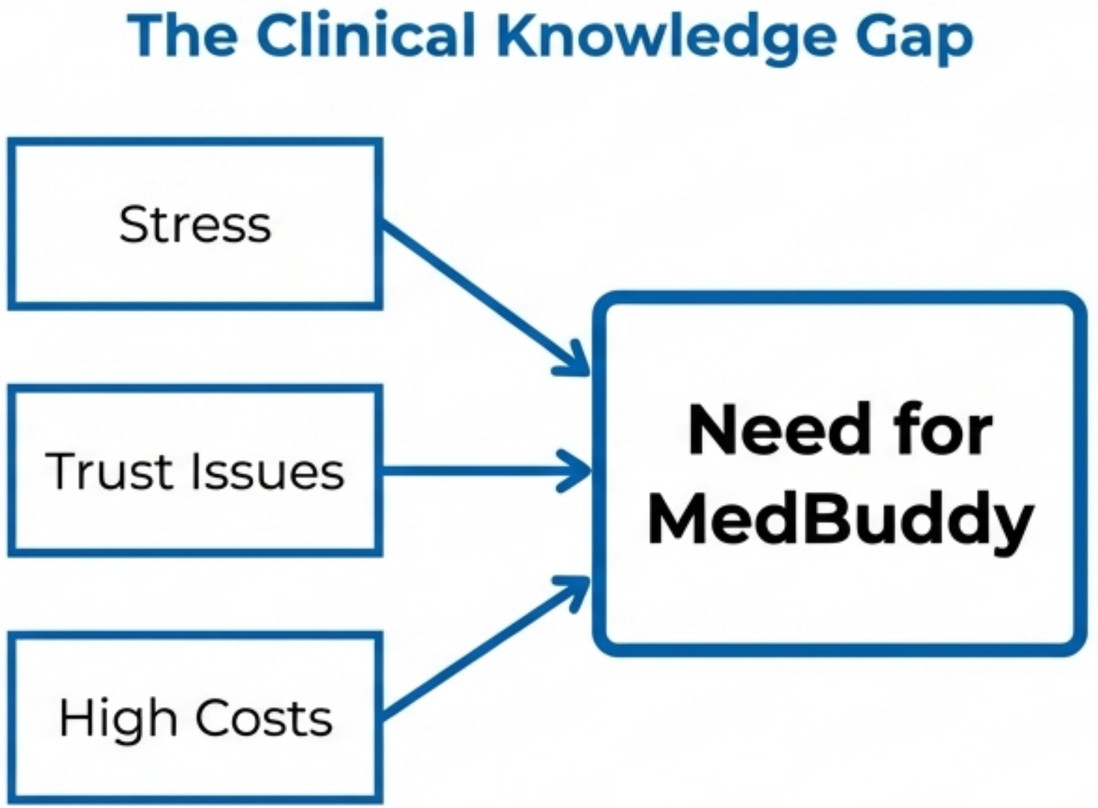
Literature Survey (Part 3)

Sl No.	Title [Year]	Author	Proposed Method	Disadvantages
7	ChEMBL Bot: Chat Bot for ChEMBL Database [2018]	V. Murali et al.	Created a chat interface to ask questions about chemical compounds and their targets.	Limited only to chemical data queries; it cannot provide guidance from clinical textbooks.

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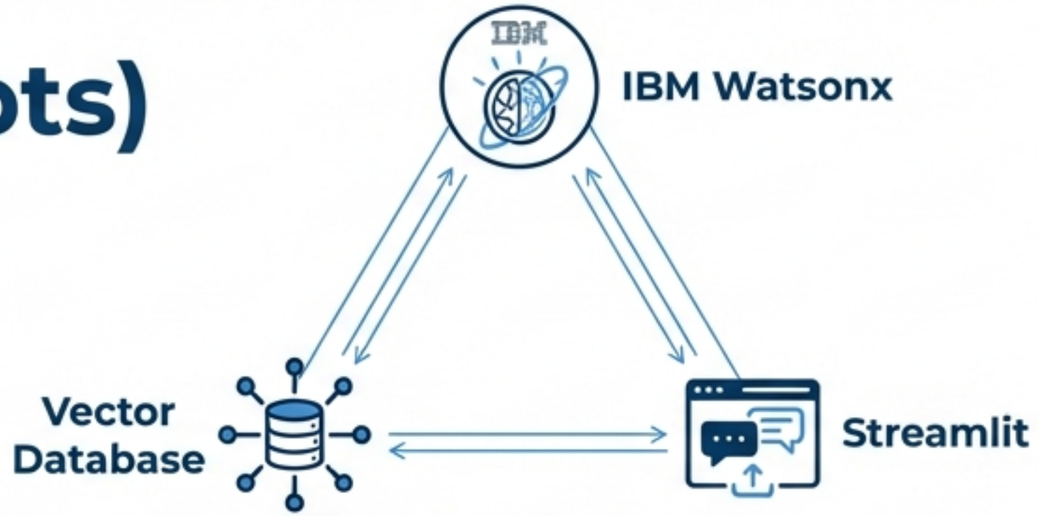
Problem Identification

- **Information Overload:** Medical students face massive workloads. Searching through 1,000+ page PDFs manually is slow and stressful.
- **The Trust Gap (Hallucinations):** Generic AI tools often guess when they don't know the answer. In medicine, a 'guess' is a safety risk.
- **Inefficient Search:** Standard 'Ctrl+F' keyword searches fail to understand the context of complex clinical questions (e.g., matching symptoms to a diagnosis).
- **Resource Inequality:** High-quality, updated medical textbooks are expensive. This limits access for students in resource-constrained areas.



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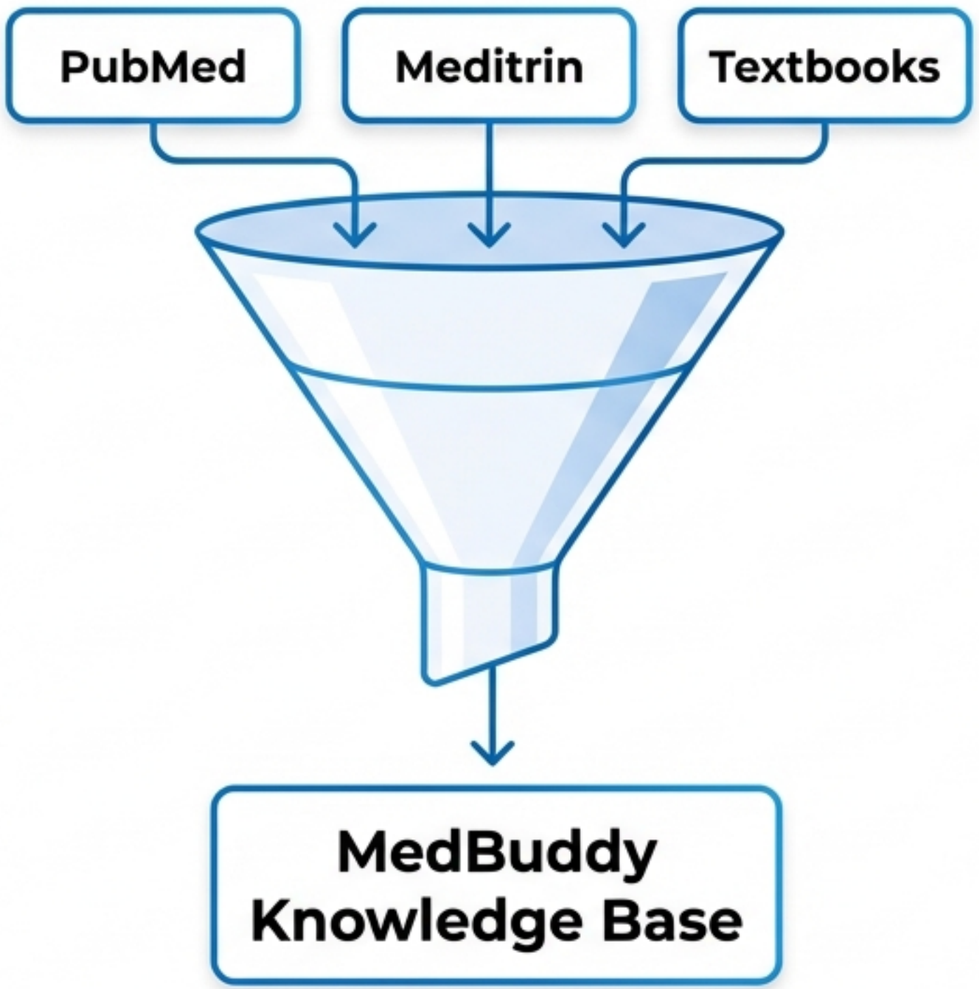
Proposed System (Concepts)



- **Embeddings**: These are the core of our search. We convert text into long lists of numbers (vectors). Text with similar meanings will have similar numbers, allowing the computer to understand concepts, not just words.
- **Vector Database**: A special storage system that holds these embeddings. It allows us to find the most relevant paragraph in a massive textbook in milliseconds.
- **Semantic Search**: Unlike a keyword search, this finds matches based on *meaning*. For example, searching 'heart attack' will also find 'myocardial infarction' because the embeddings are close together.
- **Granite-13b-chat**: The specific AI model we use from IBM. It is 'Instruction Tuned,' meaning it is trained to follow our command: 'Answer strictly using the provided context.'

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Proposed System (Datasets & Integration)



Data Integration:
We utilize **PubMed** (for research abstracts) and **Meditrin** (for clinical data) to ground the AI.

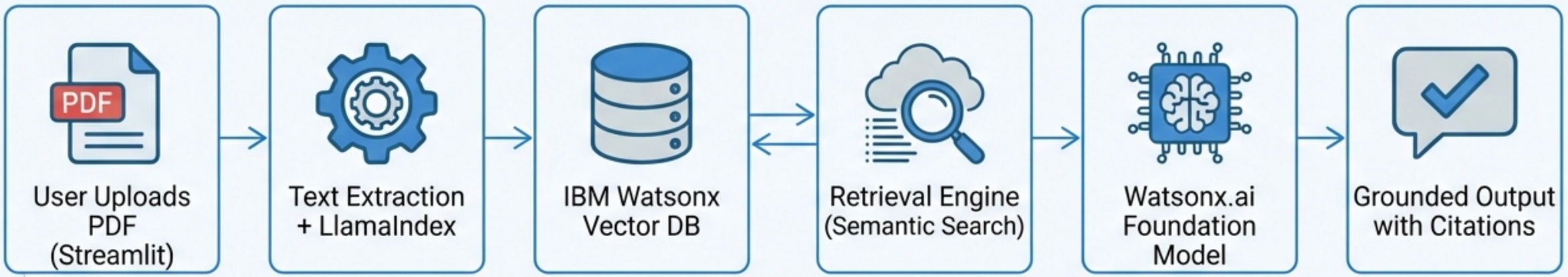
Libraries Used:

- **LlamaIndex:** A data framework that acts as the bridge between our PDF data and the AI model.
- **ChromaDB / VectorStore:** The technology used to store the mathematical map of the book.
- **HuggingFace Embeddings:** A library that provides the 'MiniLM' model to turn text into numbers efficiently on a laptop CPU.

Security: By using local secrets management, we ensure that while we use cloud intelligence, our access keys remain private and secure.

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Architecture



Input:
Student Interface

Processing:
Data Chunking & Embedding

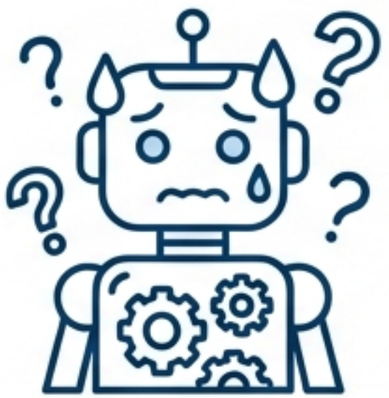
Storage:
Vector Indexing

Output:
Evidence-based answer

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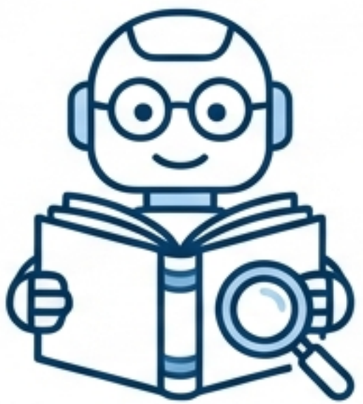
Algorithm

Closed Book
(Standard LLM)



The AI tries to memorize everything and often guesses (Hallucination).

Open Book
(MedBuddy RAG)



The AI looks up the specific page in the textbook *before* answering.

The Mechanism:

1. **Retrieve** (Fetch facts) → 2. **Context** (Summarize sections) → 3. **Answer** (Generate grounded response).

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System and Software Requirements

Software Requirements

- **Front End: Streamlit**. A pure Python framework that allows us to build the website interface without writing HTML or CSS.
- **Back End: Python 3.12**. The core programming language used for logic.
- **Libraries:**
  - **ibm-watson-machine-learning** (to connect to IBM Cloud)
  - **pypdf** (to read textbook files)
  - **python-dotenv** (for security)
  - **llama-index** (for RAG logic)

Hardware Requirements (Minimum Specs)

- **Device:** Standard Laptop.
- **CPU:** Intel Core i5 or AMD Ryzen 5 (Optimized for CPU-only inference).
- **RAM:** 8 GB Minimum (to hold the Vector Index in memory).
- **Storage:** 256 GB SSD (sufficient for storing Python environments and PDF datasets).

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Project Plan

Jan 16, 2026

Data Ingestion & Indexing

Implement the pipeline for students to upload medical PDFs and process them.

Feb 15, 2026

RAG Model Integration

Connect IBM Watsonx APIs to establish the Retrieval-Augmented Generation model.

Mar 05, 2026

UI Development

Use Streamlit to build the intuitive user interface.

Mar 15, 2026

Final Deployment

Final validation and deployment on Watsonx's secure cloud infrastructure.

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Future Scope



Clinical Decision Support

In the future, this system can act as a real-time ‘co-pilot’ for doctors, quickly looking up drug interactions or treatment guidelines.



Multimodal Capabilities

Expanding the AI to understand not just text, but also medical images like X-rays or MRI scans.



Voice Integration

Adding voice commands so surgeons or busy students can ask questions hands-free.

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Conclusion





Clinical Precision meets Modern Tech

Summary:
MedBuddy successfully bridges the gap between complex AI technology and human needs.

Democratization: It makes expensive medical knowledge accessible to everyone, regardless of their location or financial resources.

Impact: By reducing search time by **70%** and achieving an accuracy (groundedness) score of **84%**, MedBuddy creates a less stressful and more effective learning environment for future doctors.

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1. V. Murali et al., “ChEMBL Bot - A Chat Bot for ChEMBL Database,” 2018 14th International Conference on Information Processing (ICINPRO), 2018.
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3. R. S. Nambiar and D. Gupta, “Dedicated Farm-Haystack Question Answering System for Pregnant Women and Neonates,” 2022 IEEE Confluence, 2022.
4. Peri, S. D. B. et al., “Chatbot to chat with medical books using Retrieval-Augmented Generation Model,” 2024 IEEE ICINPRO, 2024.
5. Baglivo, F. et al., “Exploring the Possible Use of AI Chatbots in Public Health Education,” JMIR Publications, 2023.

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